

Scalability & Future Upgrade Strategies

Smart-Lender Project Documentation Suite

Project Name	Smart Lender - AI-Powered Loan Approval Prediction System
Document Type	Scalability & Future Upgrade Strategies
Project Phase	Phase 8: Project Demonstration
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Organization / College	Vishnu Institute of Technology
Status / Version	Approved / v1.0 (Exact Replica Format)

1. Scalability Goals

Scalability Dimension	Observation Limitation	Future Proposed Architecture Plan
Database Storage	Stateless Flask scripts (no backup data history)	Implement PostgreSQL/MongoDB to maintain audit trail archives
User Authentication	No access roles (Anyone can view system portal)	Incorporate OAuth2 structures and roles mapping controls
Model Operations	Static joblib storage folder configuration	Migrate model artifacts folder to dynamic AWS SageMaker endpoints
API Decoupling	Server scripts run html template renderings	Decouple frontend layout from backend APIs using React & FastAPIs

2. Future Project Major Upgrade Roadmap

Project Phase	Upgrade Category Target Details	Timeline Schedule
Phase 2 Upgrade	Decouple into logical React app and independent microserver API routes	Month 1-2
Phase 3 Upgrade	Establish databases for user registration and loan logging metrics	Month 3
Phase 4 Upgrade	Connect third-party banking APIs to fetch verified statements in real time	Month 4-5